

Medical Affairs, work redesigned

AI in Action · October 13–14, 2026
Convene Philadelphia

Vivek Mukhatyar

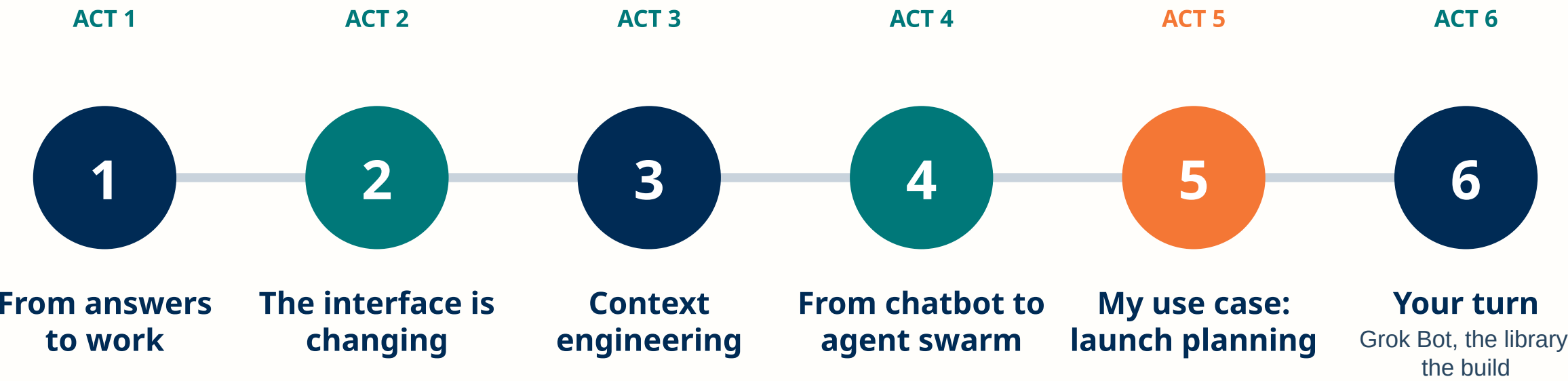


DISCLOSURE

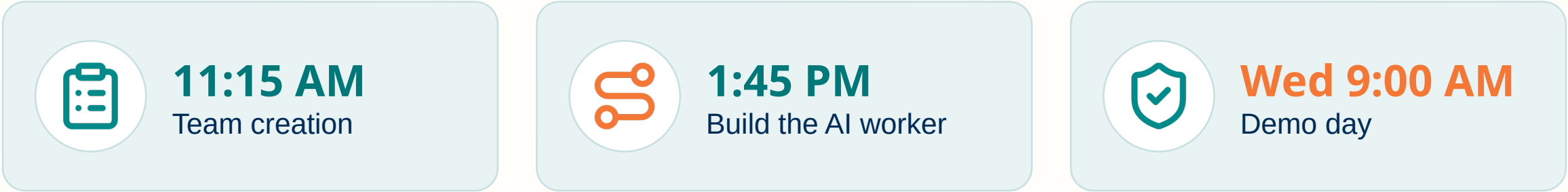
The views presented here are my own and do not reflect those of my current employer.

Vivek Mukhatyar

Agenda: six acts, then we build



THEN, HANDS-ON



Four objectives for today

01

Give AI goals, not questions

Move from summarizing and finding answers to orchestrating and executing whole workflows



02

Meet the frontier agents

Claude.ai, Microsoft Copilot, ChatGPT, Grok Bot, Meta Muse: your own personal workforce



03

Context is our secret sauce

Prompt engineering becomes context engineering. Medical Affairs holds the context.



04

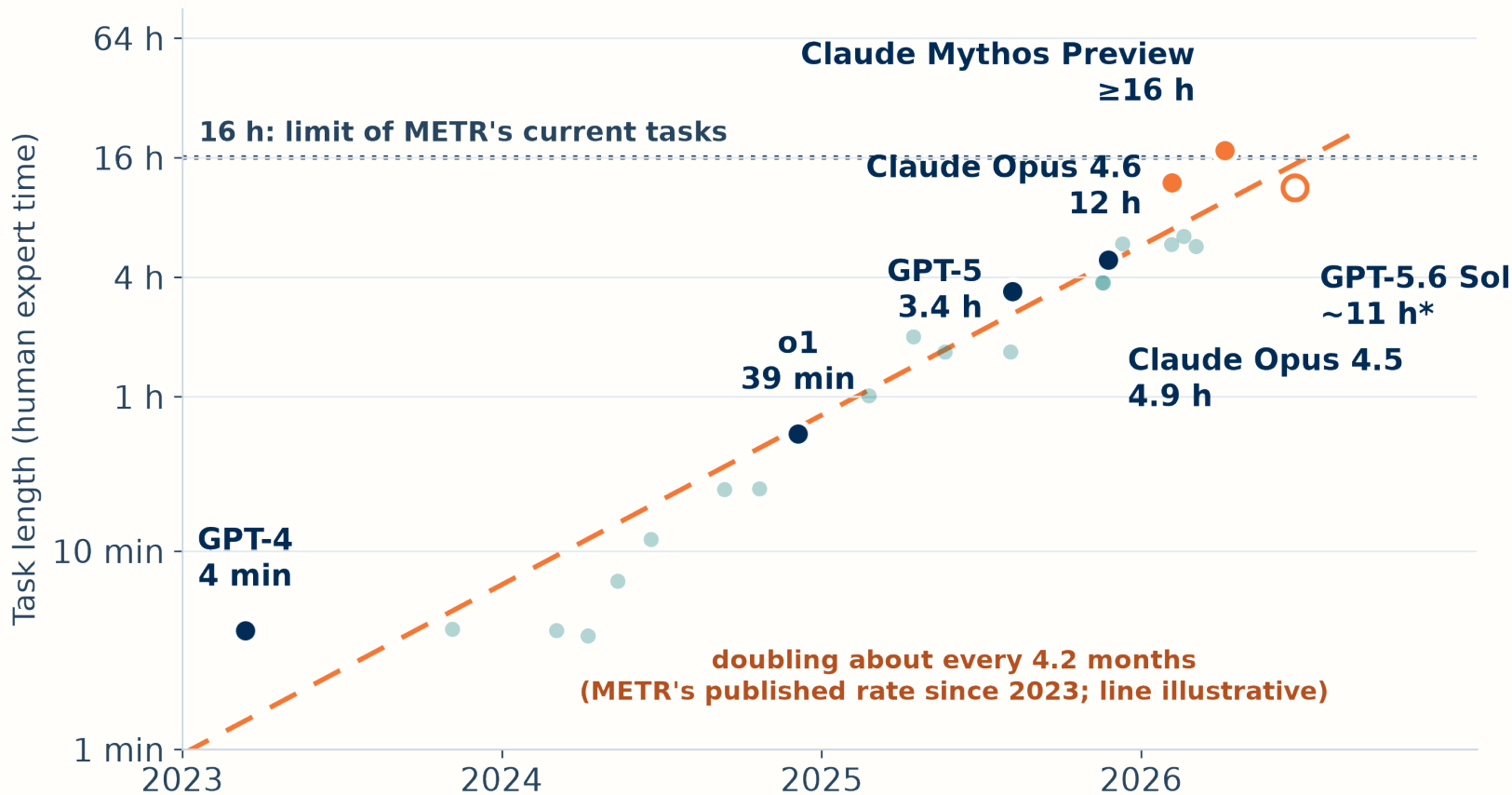
Have fun building

Form a team, build a real AI worker together, and show it on demo day



AI agents went from minutes-long tasks to day-long tasks in three years

METR 50% time horizon: about 4 min for GPT-4 (March 2023) → 12 h for Claude Opus 4.6 and at least 16 h for Claude Mythos Preview (2026)



SEPTEMBER 2026 FRONTIER too new for METR

Epoch Capabilities Index (ECI) index points, independent	
Claude Opus 5.5	167.3
GPT-6 Astra	166.5
Grok 4.7	153.5

Cursor Composer 2.5 May 2026, Cursor-reported	
CursorBench 4.0	27.7%
Terminal-Bench 2.0	69.3%

Different units: scores, not hours. Not plotted on the chart.

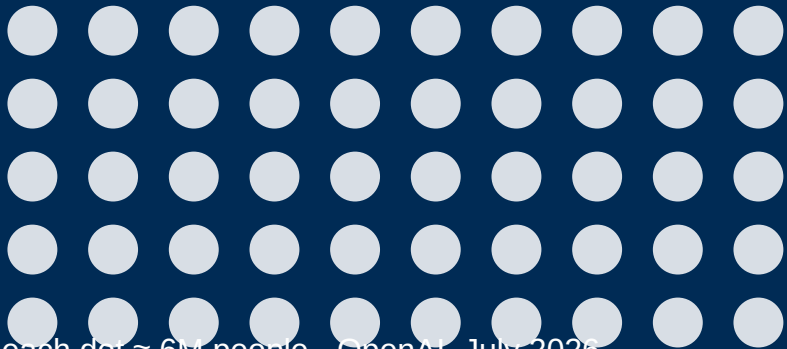
Task length AI can complete has doubled about every 4 months since 2023, and the newest models have outgrown the test.

METR time-horizons (updated May 2026) and METR posts on X; Epoch AI Capabilities Index; CursorBench / Cursor. 50% success on well-defined software tasks, measured by how long a human expert takes. Doubling: METR 128.7 days since 2023 (95% CI 104–158). Top points have wide ranges (Opus 4.6: 5–61 h; Mythos: 8–55 h). *GPT-5.6 Sol: pre-deployment, METR post on X.

Scientific questions already reach AI at scale

300M+

people a week ask ChatGPT health questions



1M+

questions a day answered by OpenEvidence



Different populations and units. Both change where scientific exchange begins.

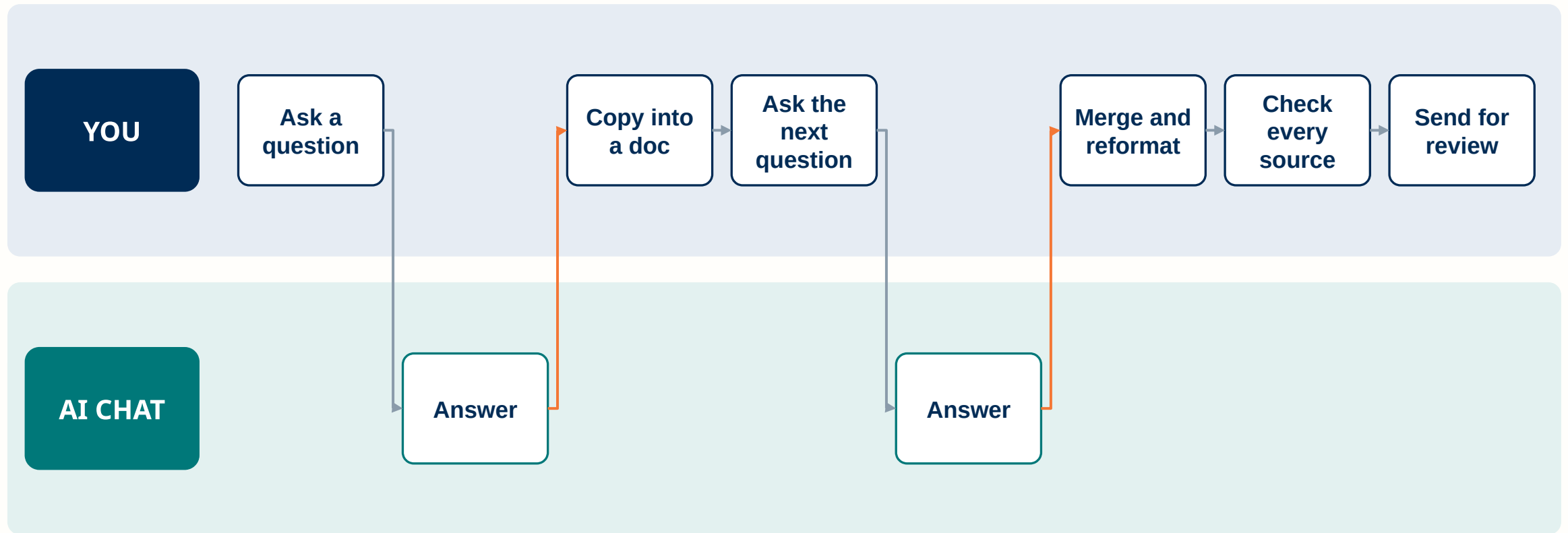
The north star remains the same

Patient outcomes

via evidence · exchange · insights



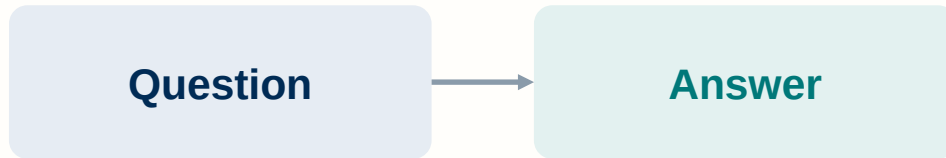
Most of us start here: one answer at a time



The human carries every handoff. The orange arrows are you, picking the work back up.

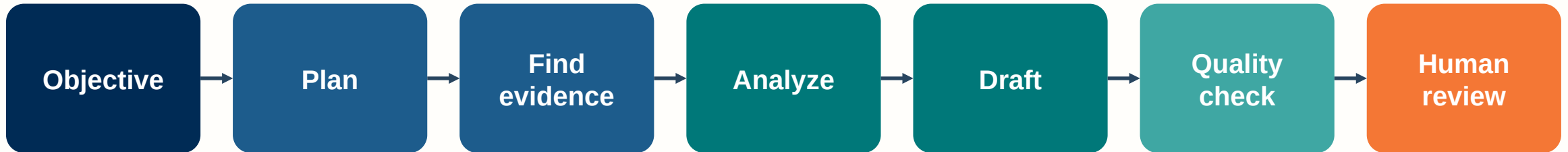
The problem is not chat. It is the unit of work.

CHAT



One exchange. Then you start over.

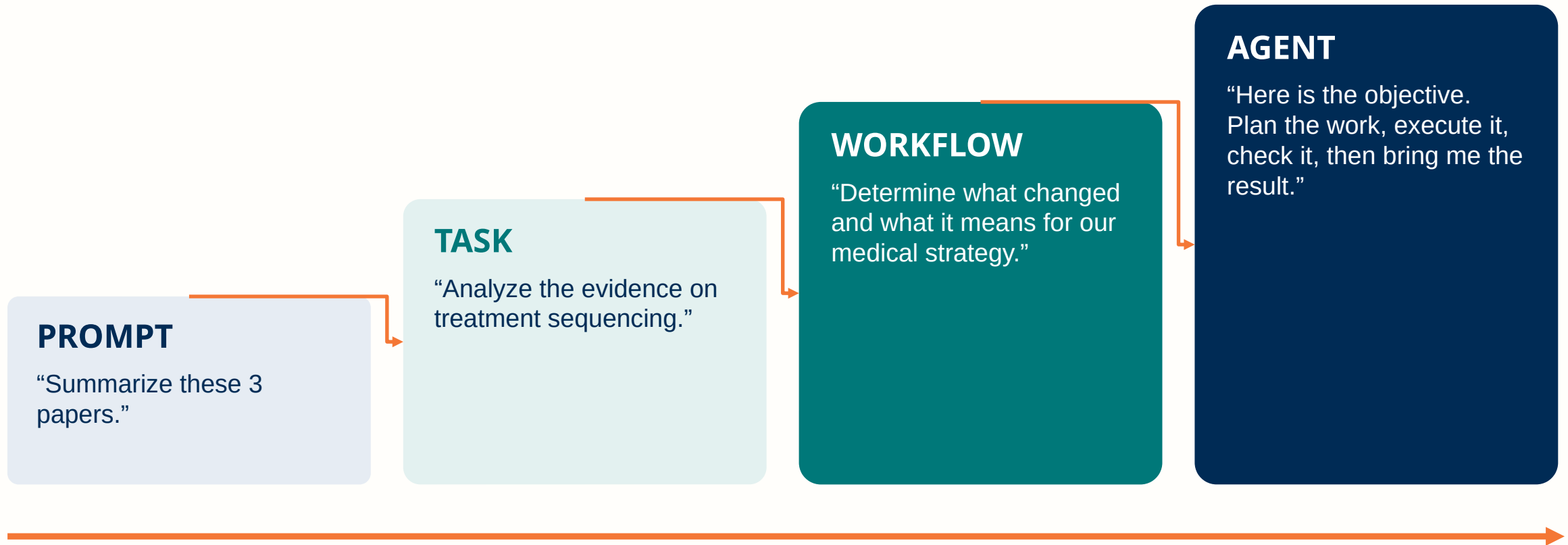
WORKFLOW



Many steps, one outcome, checked before it reaches you

Stop optimizing the answer. Start designing the work.

Prompt → task → workflow → agent



From prompts to assignments

A PROMPT

“Summarize obesity papers”

One answer. Repeated handoffs.



AN ASSIGNMENT

Goal

Reviewable ad-board package: sustaining treatment

Context

Our evidence position, house rules, past questions

Sources

PubMed, ClinicalTrials.gov, approved inputs

Output

Charter, agenda, pre-read, claim ledger

Checks

Every claim cited; limitations stated

Review

A named human approves before use

Stop asking AI questions.

Start giving it jobs.

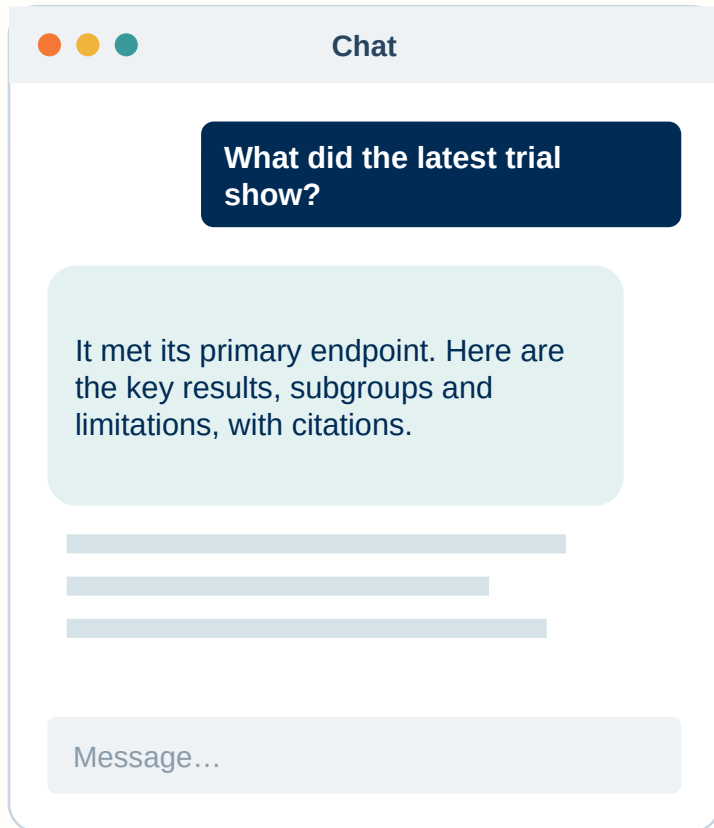
ACT 2

The interface is changing

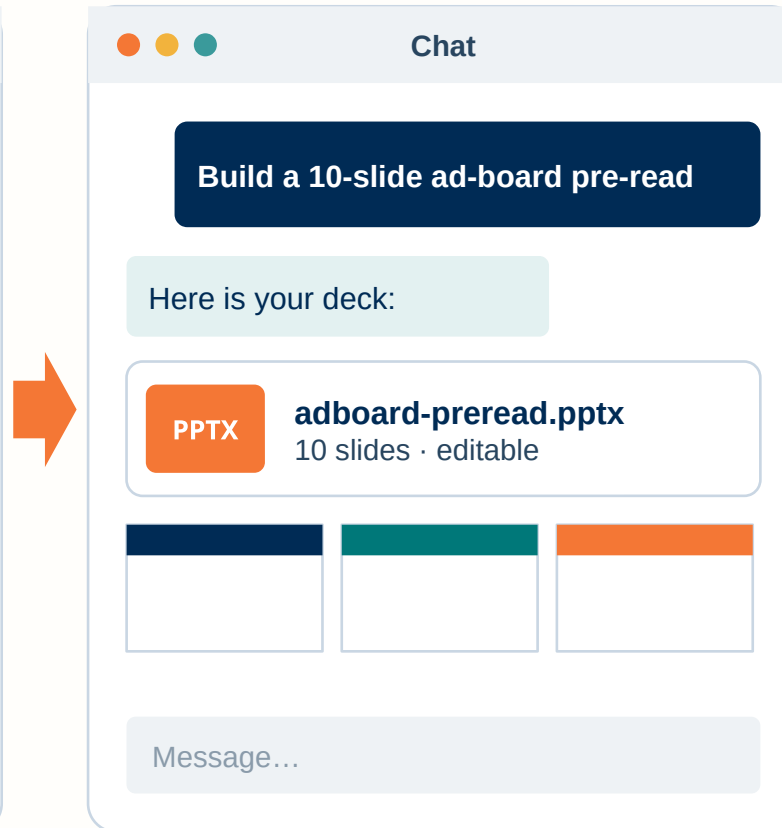


The interface changed: same window, different job

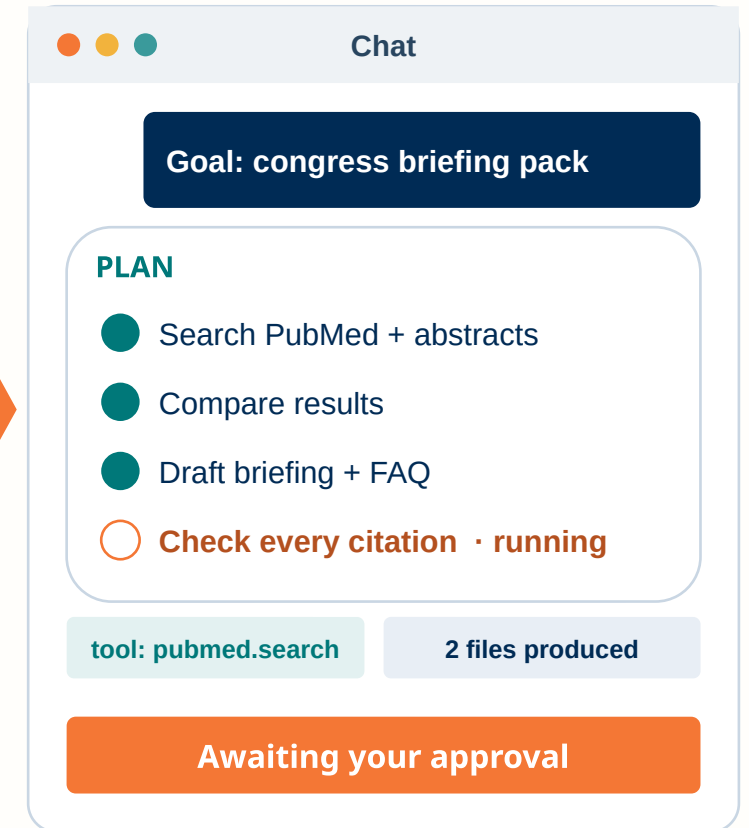
1 · ANSWERS



2 · ARTIFACTS

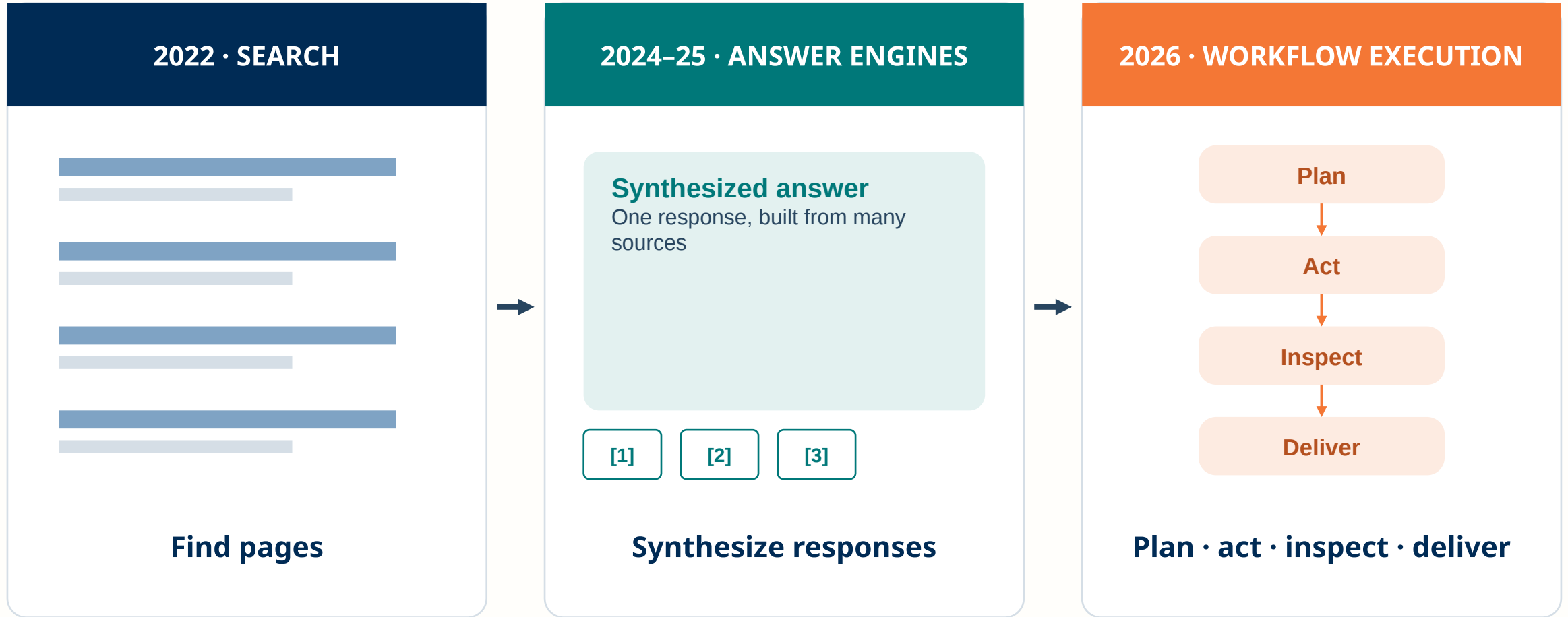


3 · AGENTS DOING WORK



Answers → artifacts → agents doing work. You move from asking to approving.

From finding information to completing work



Meet the frontier agents



Claude.ai

Anthropic



**Microsoft
Copilot**

Microsoft



ChatGPT

OpenAI



Grok Bot

xAI



Meta Muse

Meta

CHAT TOOLS
You ask, it answers

the shift

PERSONAL AGENTS
Your own workforce

Same assignment and checks · different access and controls

Your own workforce: a small team, clear handoffs

YOU · set the goal, approve the result

Coordinator

Owens the outcome

Evidence worker

Returns sources + limitations

Design worker

Returns decision map + draft

Reviewer

Returns defects + severity



Every handoff carries inputs, sources, open questions and a done-check

The new vocabulary of AI

Model

The reasoning engine that reads and writes language

Agent

A model pursuing a goal through tools and feedback

Harness

The environment that manages tools, state, permissions and checks

Context

Everything the model can see for this step

Memory

Saved information and progress across steps

Skills

Reusable instructions for a type of work

Tools

Permitted ways to reach a source or take an action

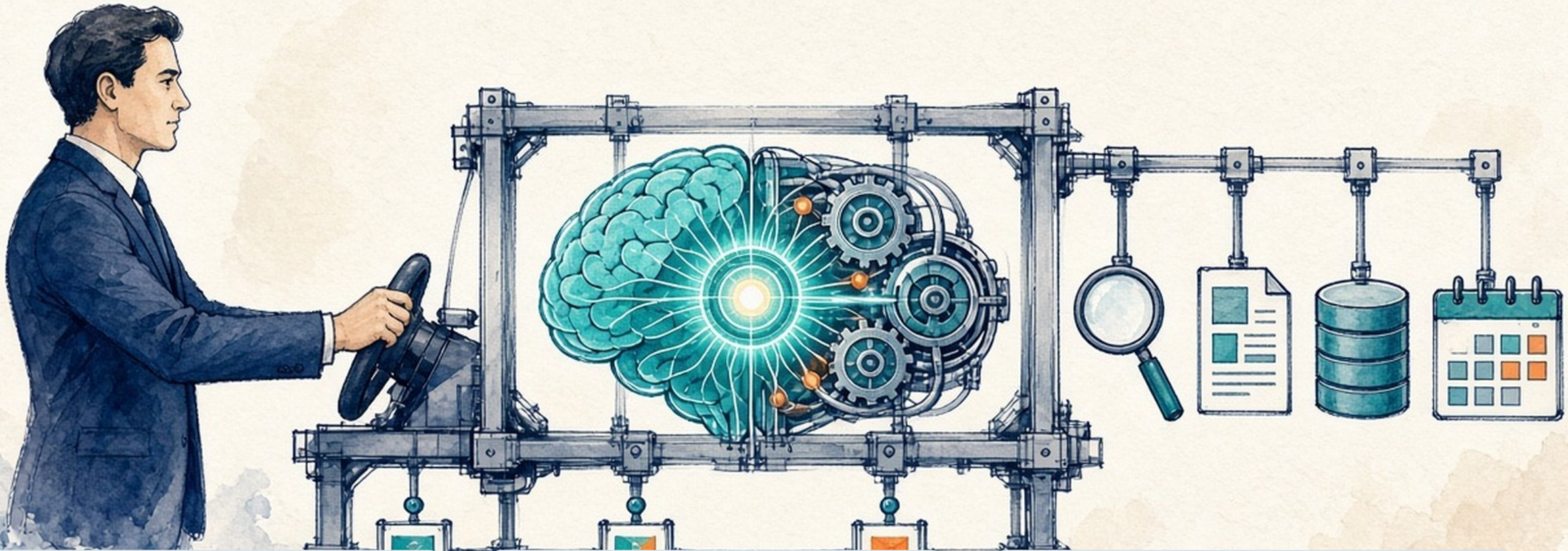
Swarm

Several specialists coordinated around one outcome

Long horizon

Work that runs across many steps and hours

What is an AI agent?



HUMAN

Purpose · boundaries · review

MODEL

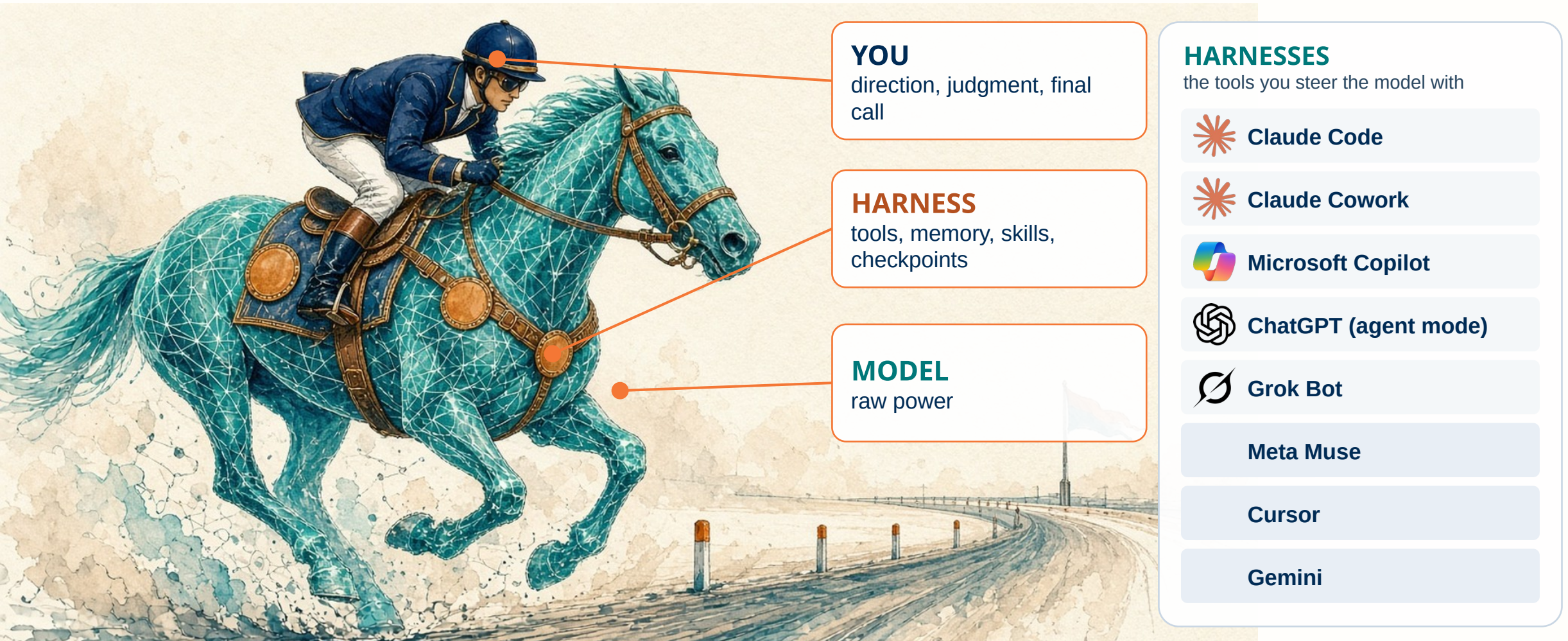
Reasoning + language

HARNESS

Tools · memory · permissions · checks

An agent = a model + a harness + human direction, working on a real job

The model is the horse. You are the jockey. The harness is the tool.



Without a harness, even the best model can't be steered through a long-horizon task.

Context is the secret sauce, and Medical Affairs holds it



Prompt

How you ask

Instructions + output format

Context

What it can see

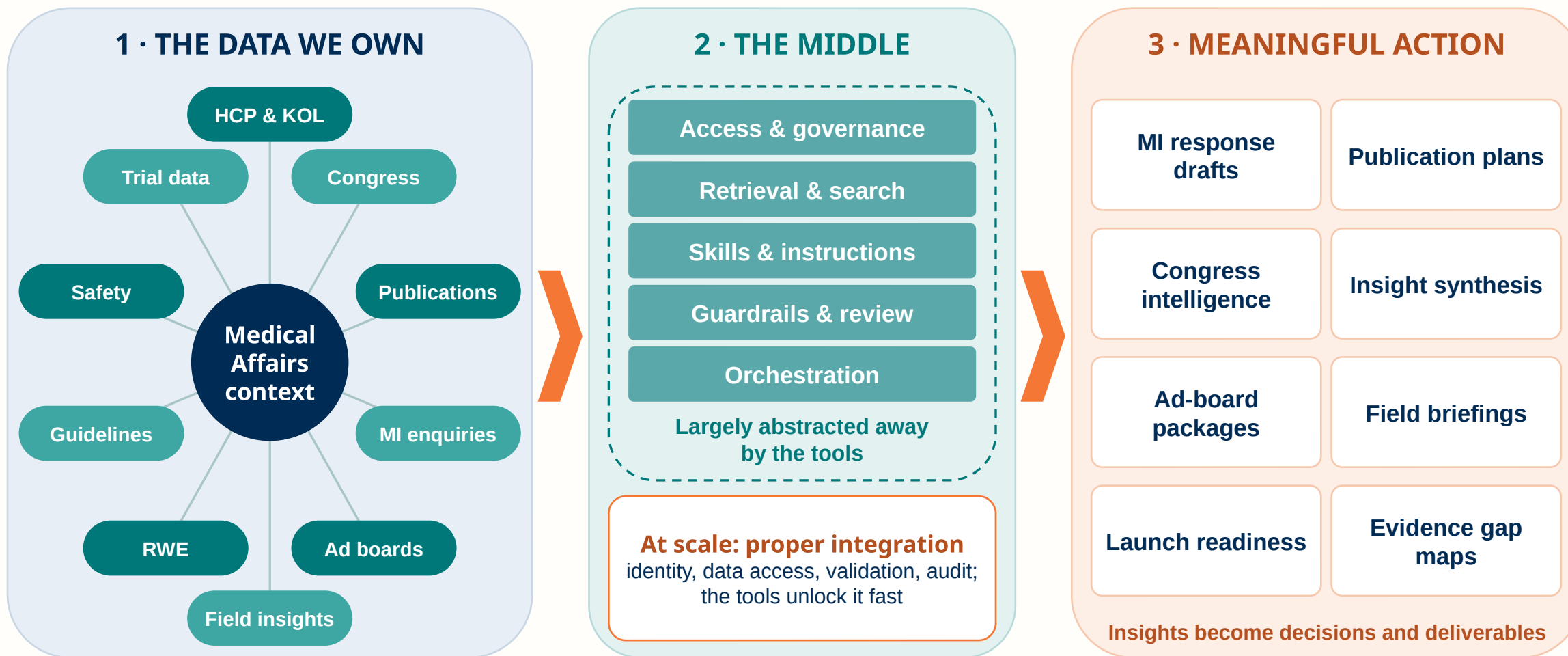
Held by Medical Affairs: evidence, house rules, past decisions, field insight

Harness

How the work runs

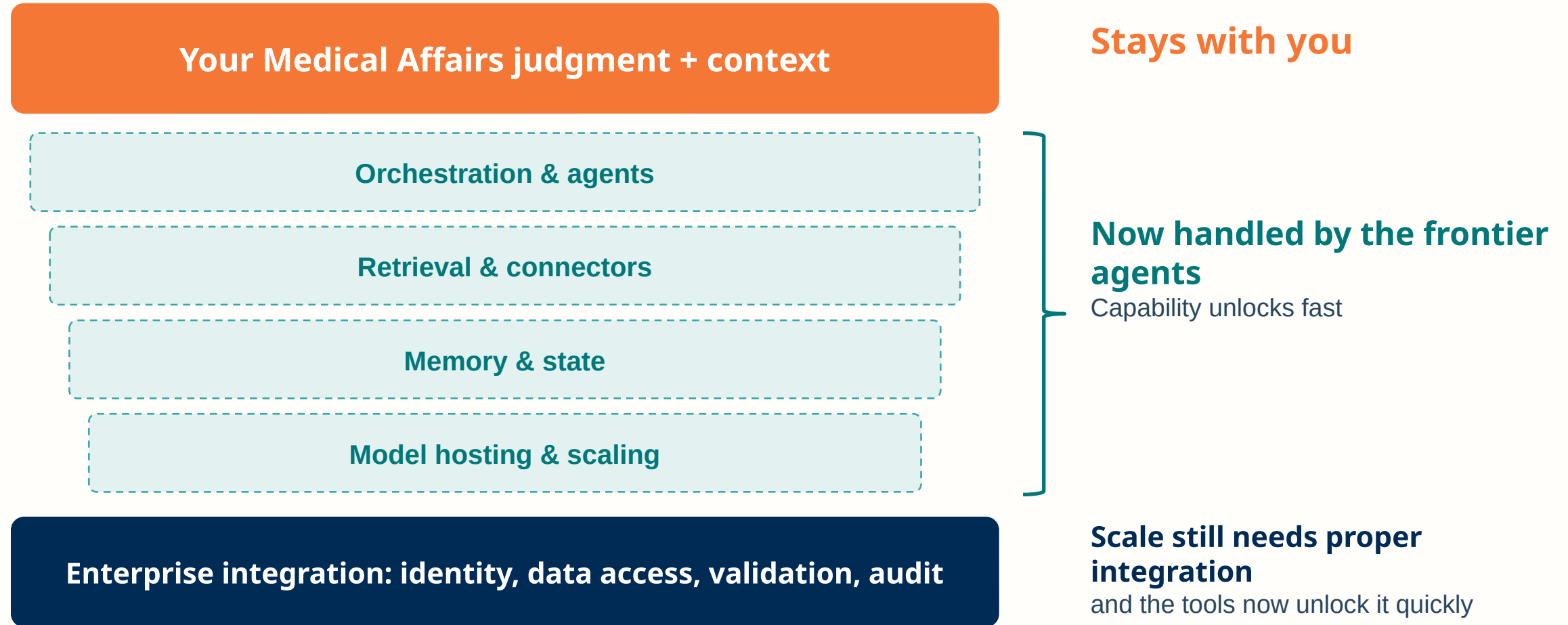
Tools + permissions + checkpoints

From the data we own to meaningful action

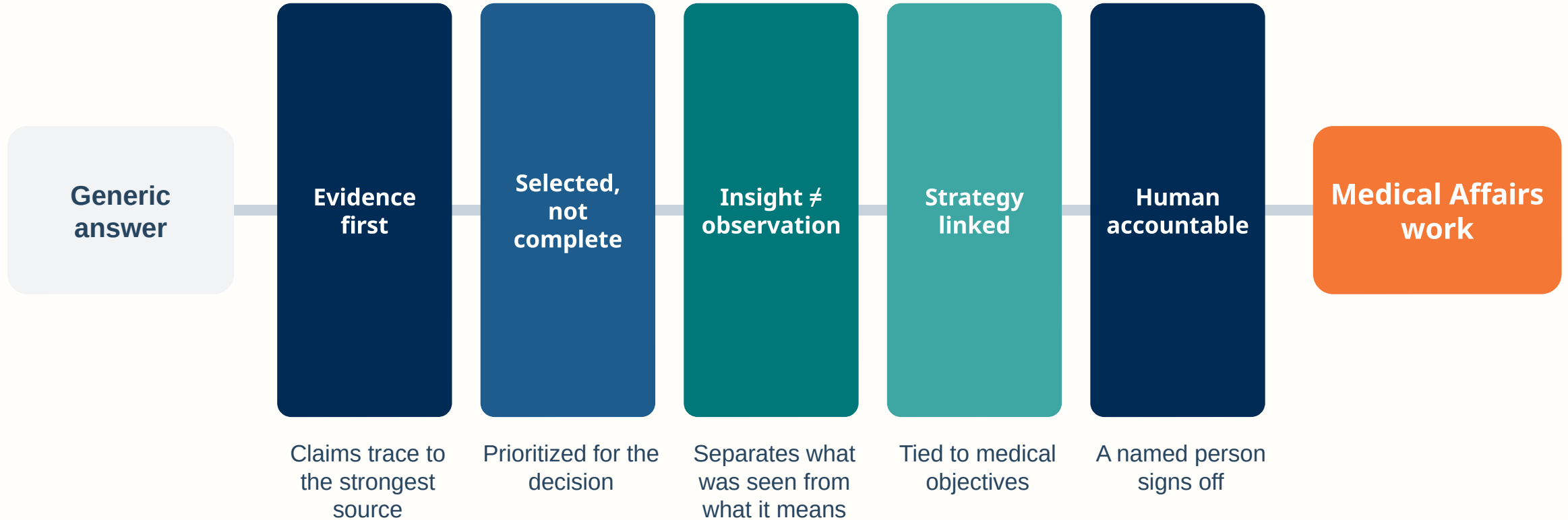


We own both ends: the context going in and the judgment on what comes out.

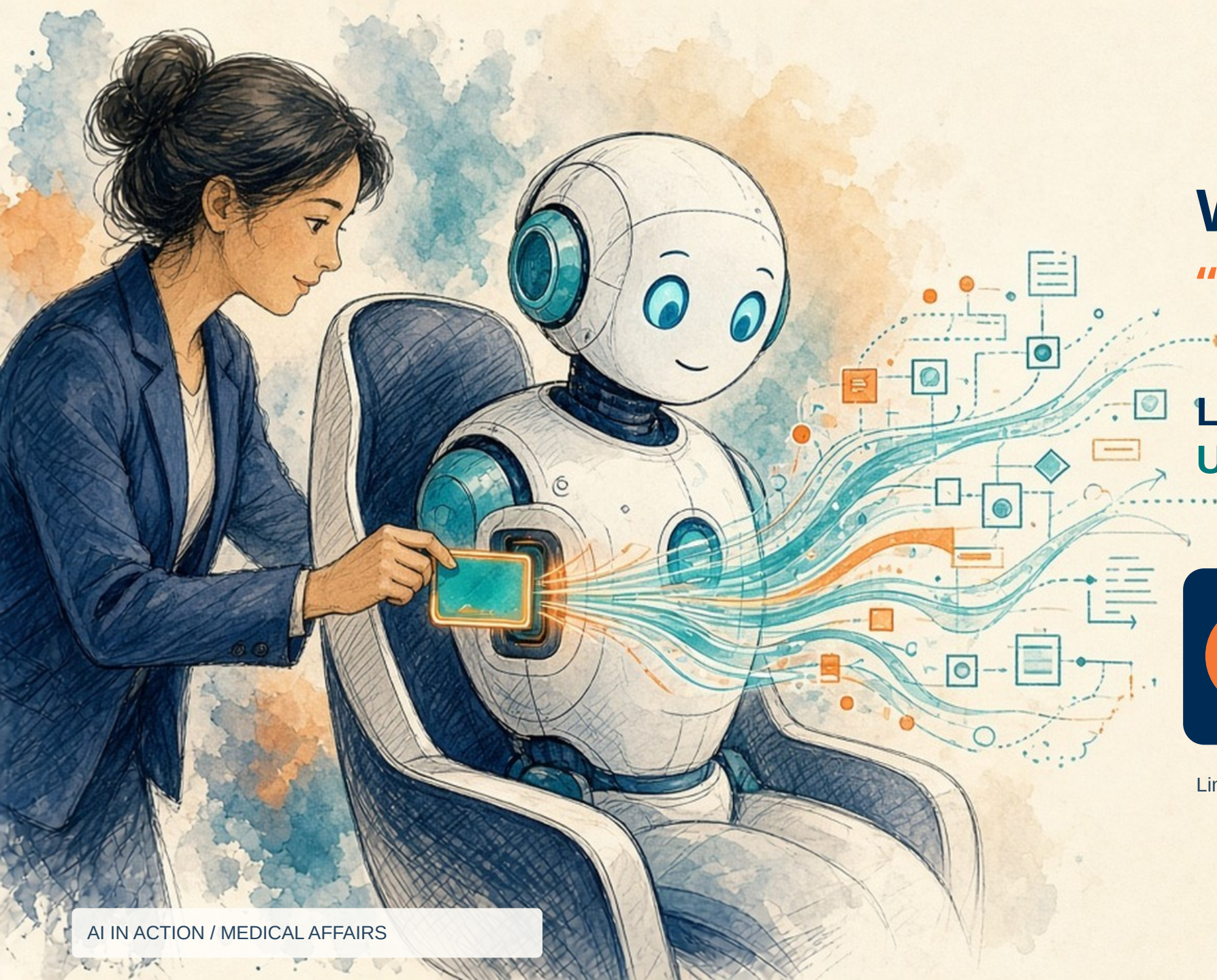
The middle is being abstracted away



What makes it Medical Affairs: encoded judgment



You are not just using AI. You are teaching it how your function thinks.



What is a skill?

"I know Kung Fu."

**Load the skill.
Use it instantly.**



Watch the clip

The Matrix 25th Anniversary · I
Know Kung Fu · Warner Bros.
(YouTube)

Link only: youtu.be/ozS_6vqnJ8w

Agent Skills make expertise reusable

GitHub is the shared shelf
A skill is the work recipe

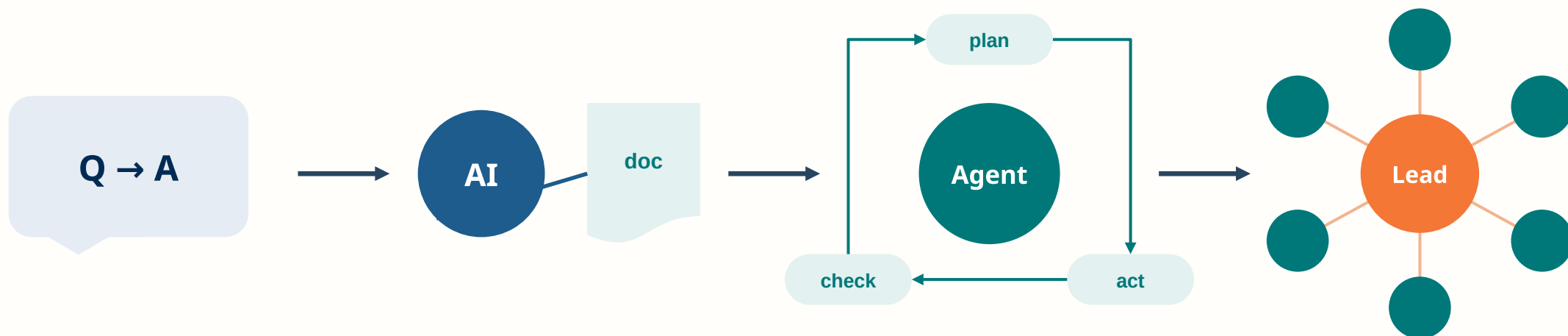
— SKILL.md → procedure

— Supporting files → examples

— House rules → boundaries



From chatbot to agent swarm



CHATBOT

Give: A question
Get: An answer
You do every step

ASSISTANT

Give: A task + some files
Get: A draft of one deliverable
You stitch drafts together

AGENT

Give: A goal + skills + house rules
Get: A finished workflow with provenance
You review and decide

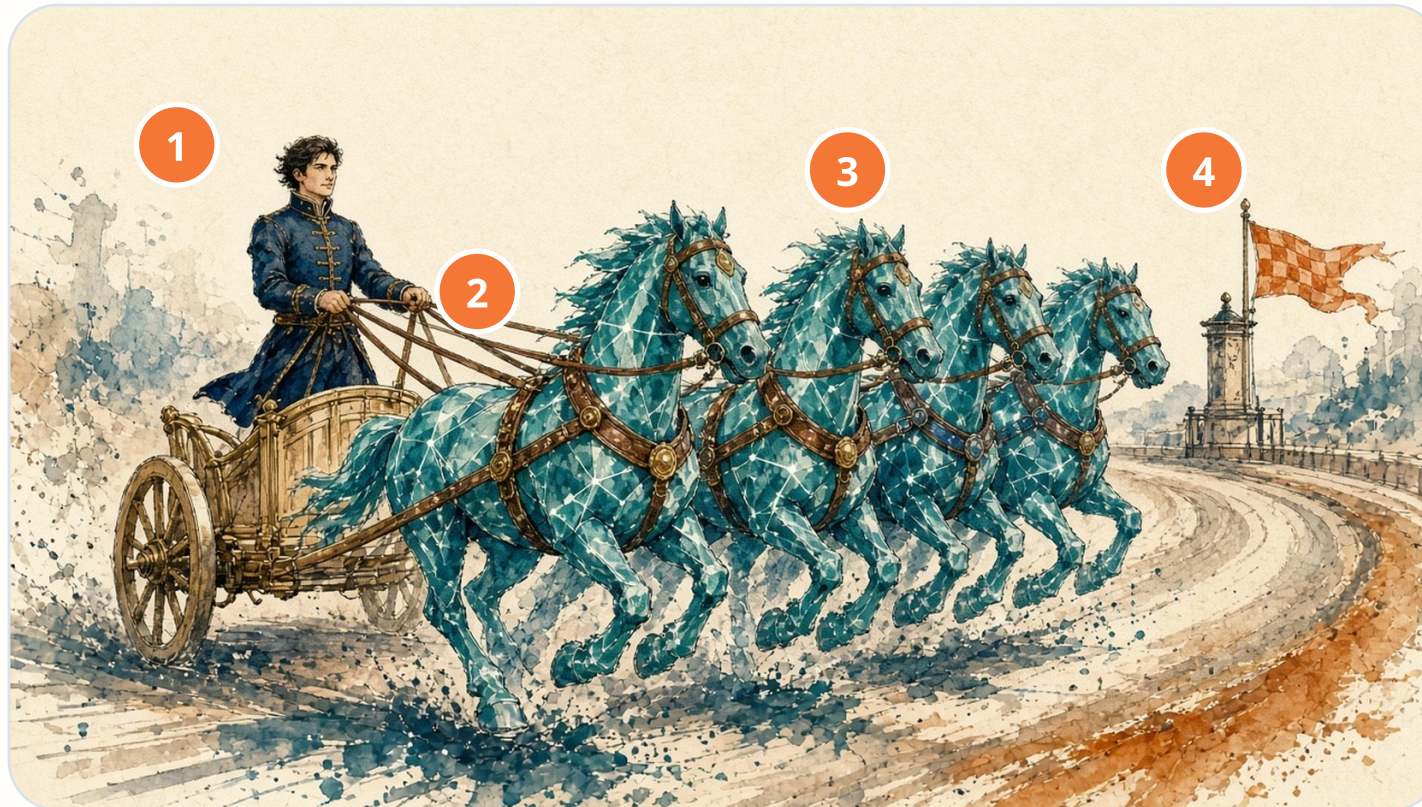
AGENT SWARM

Give: A goal + an org chart + a context packet per role
Get: A coordinated plan, an auditor, human gates
You lead and judge

The jump to a swarm is structure, not a bigger prompt: roles, context, handoffs, gates

What is an agent swarm?

A team of AI agents, each with its own role, skills and context. A lead agent splits the job, hands work between them, merges the results and stops at human checkpoints.



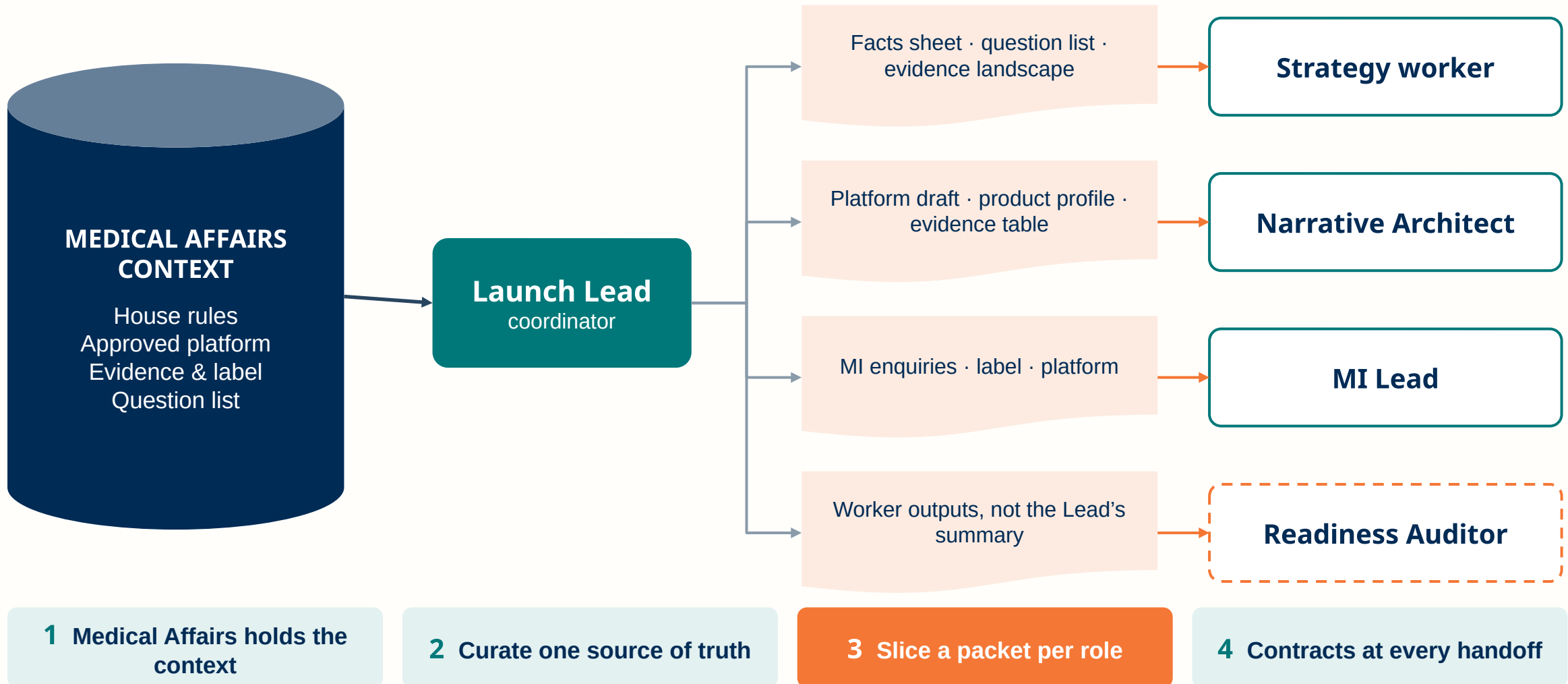
ONE HUMAN, MANY HORSES

- 1 Charioteer = you**
the human at the helm sets the direction
- 2 Reins + harness = lead agent + rules**
splits the job and hands work along
- 3 Each horse = a specialist agent**
its own skill and context
- 4 Finish post = human checkpoint**
nothing crosses without your decision

MODEL the horse → **HARNES** tools + rules →
AGENT + a job → **SWARM** many horses

One agent for a single task. **A swarm** for multi-part work with hand-offs, like a launch plan.

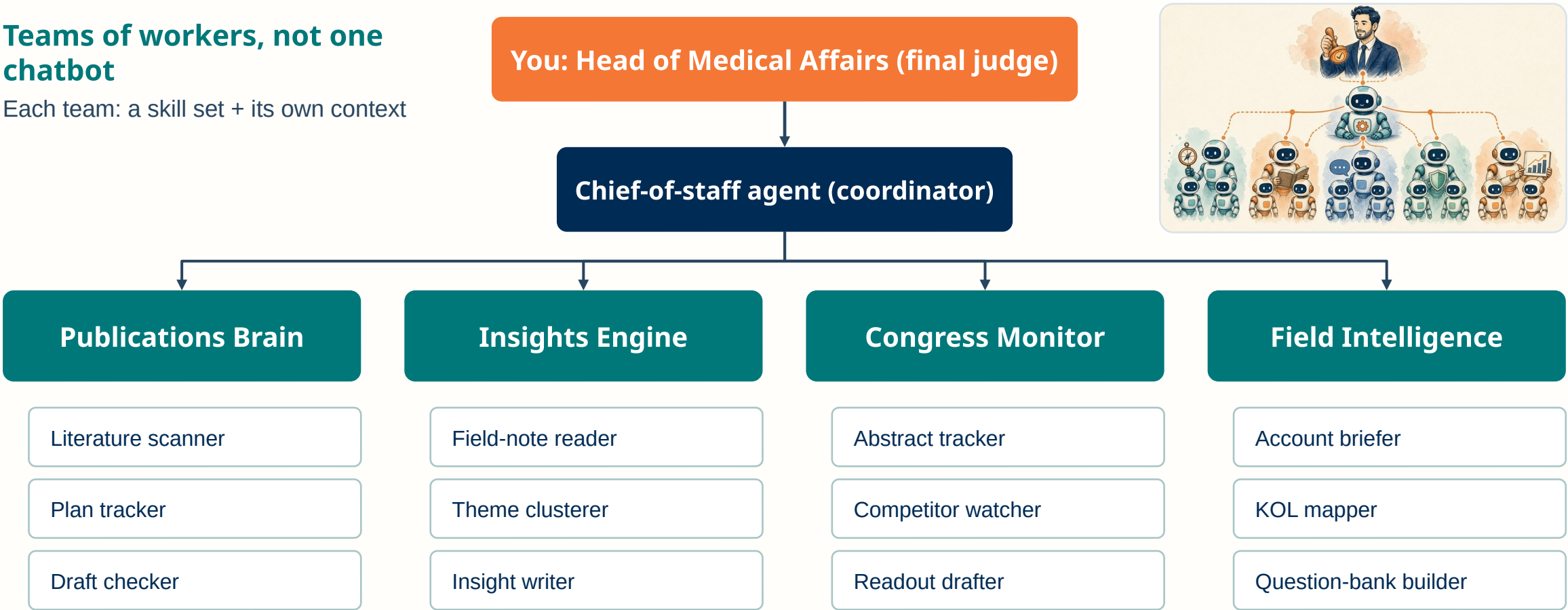
Context engineering feeds every sub-agent



Your org chart of agents

Teams of workers, not one chatbot

Each team: a skill set + its own context



Digital workers you assign, bound and review: you stay at the top of the chart

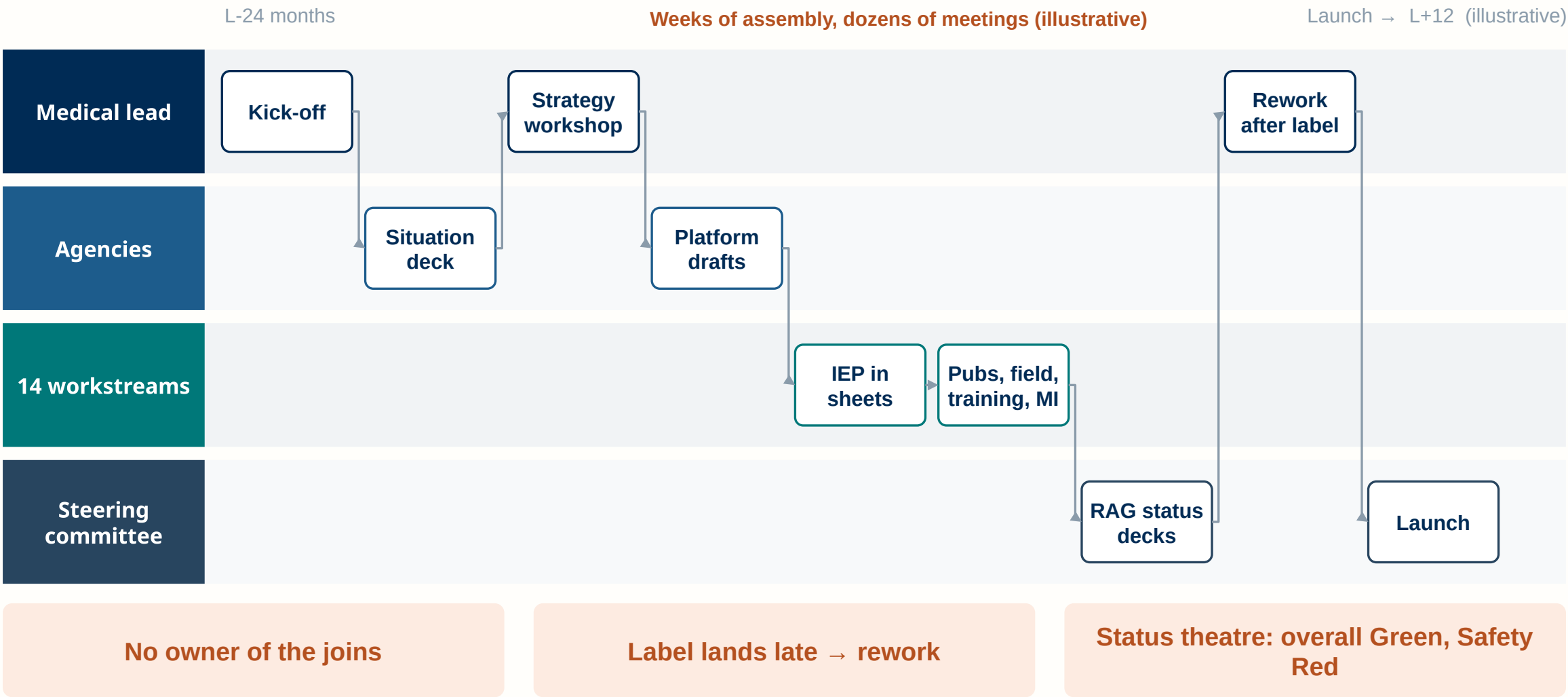
ACT 5

My use case: the whole launch plan

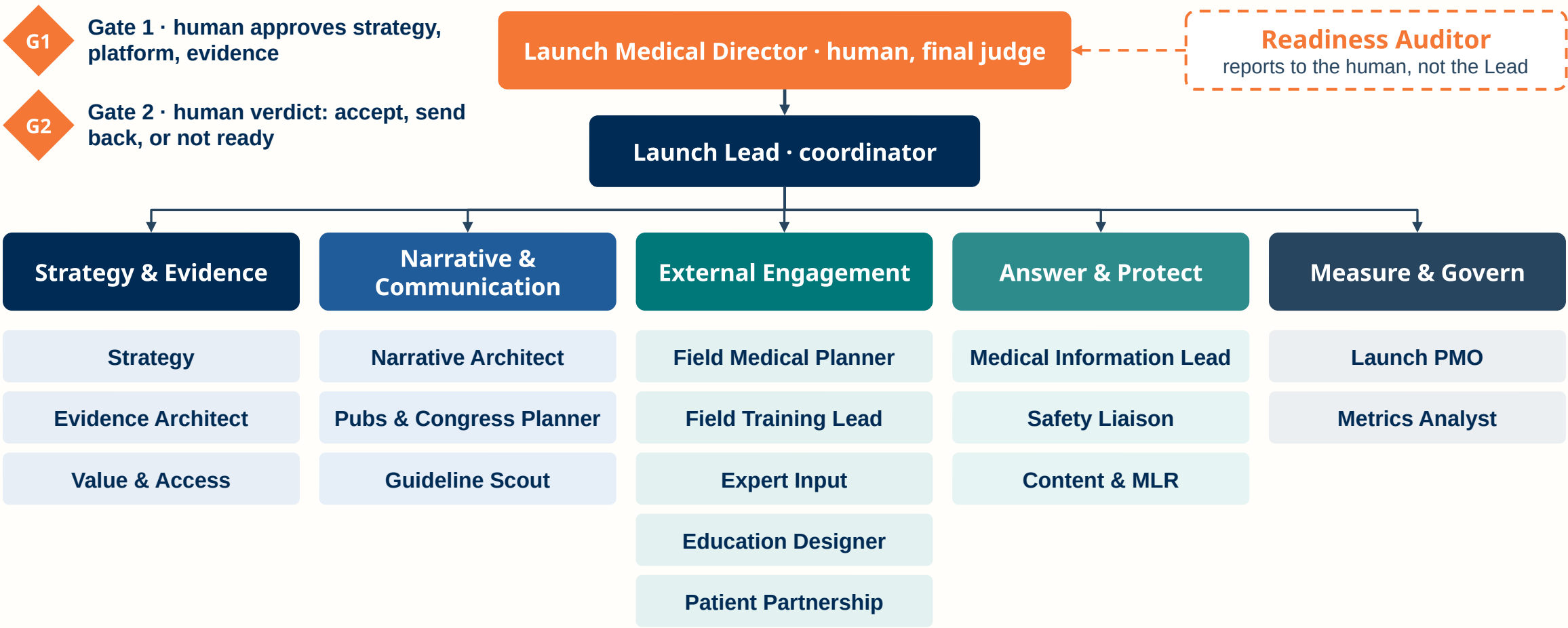
A team of digital workers builds it.
A human decides.



Today: fourteen decks, no owner of the joins

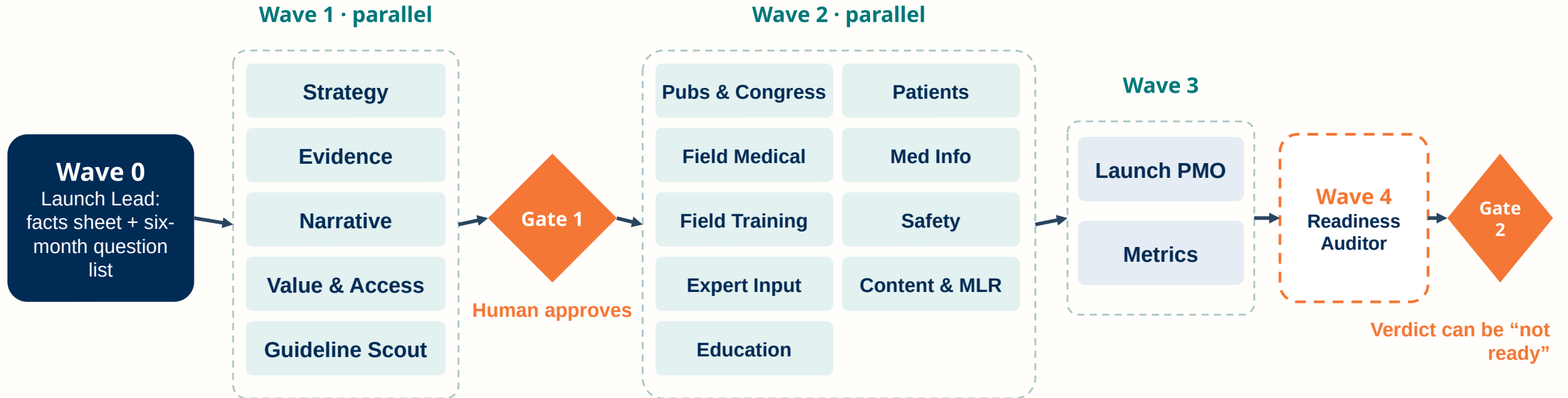


The launch plan swarm: an org chart of digital workers



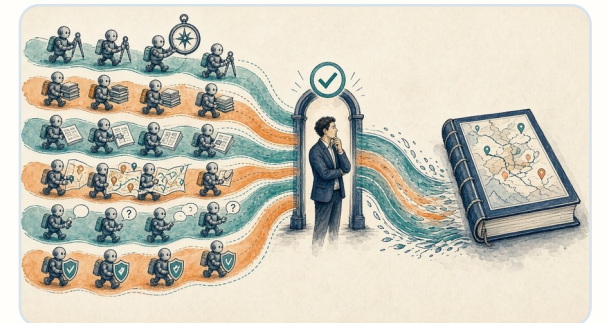
16 specialists · 5 pods · 1 coordinator · 1 auditor · 2 gates · a human is the final judge

How it runs: waves and human gates



Wave 1 sets the strategy and the words; a human approves them at Gate 1.

Wave 2 runs in parallel on the approved platform. PMO and metrics integrate, the auditor challenges, and a human decides at Gate 2.



The human is always the final judge

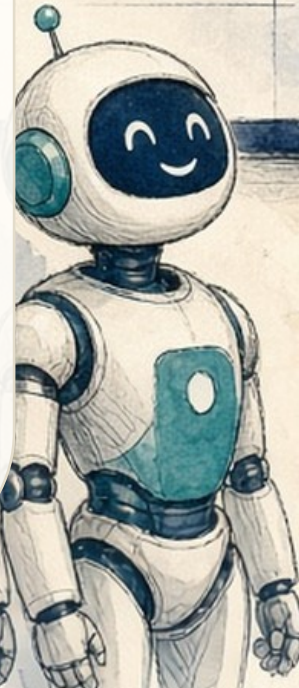
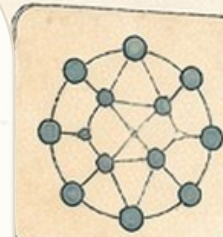
Agents draft. People decide.

Every gap has a named owner and a date.

Gate 1 · approve strategy + platform

MLR + Safety · certify; handle real cases

Gate 2 · accept, send back, or not ready



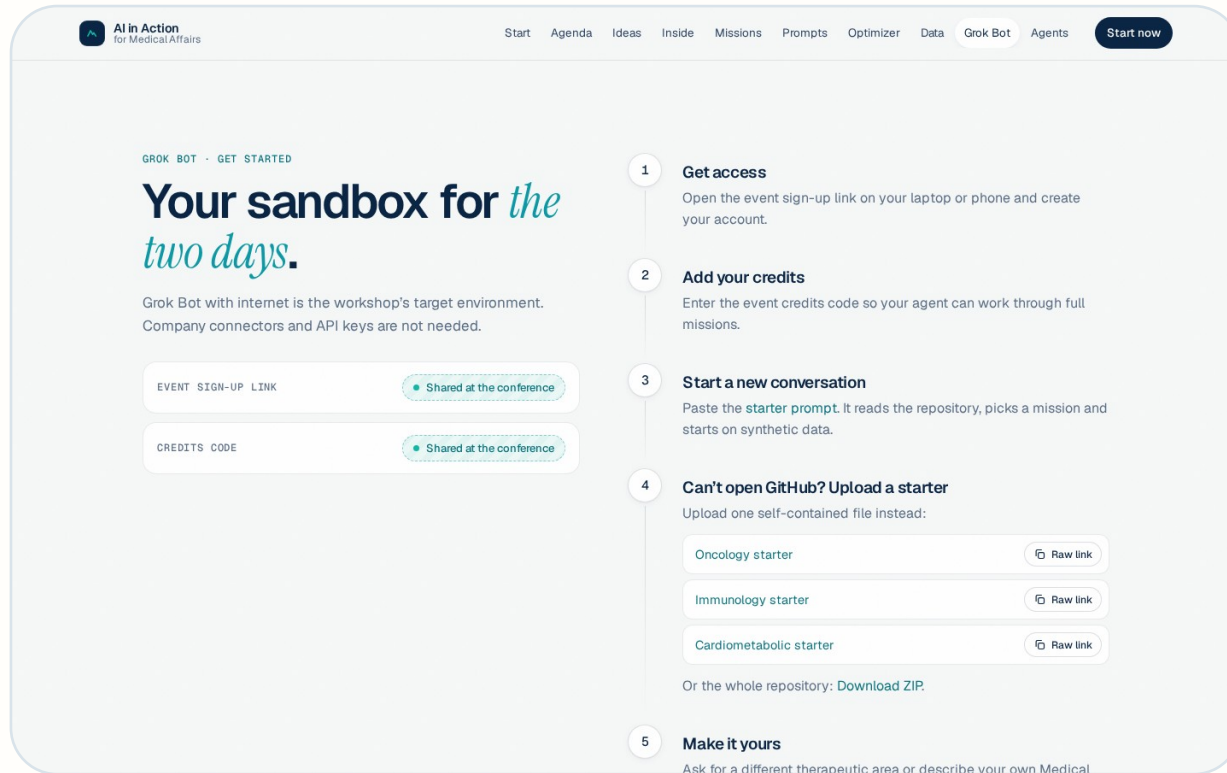
Autonomy lives inside a safe lane



Widen the lane only after the evidence says you can

Get Grok Bot running in four steps

Grok Bot with internet is the workshop sandbox. No company connectors or API keys needed.



Companion website · Grok Bot page (preview)

- 1 Get access**
Open the event sign-up link and create your account.
- 2 Add your credits**
Enter the event credits code so your agent can run full missions.
- 3 Start a new conversation**
Paste the starter prompt. It reads the library, picks a mission and starts on synthetic data.
- 4 Make it yours**
Ask for another therapeutic area, or describe your own Medical Affairs goal.

Sign-up link and credits code: shared at the conference · Can't open GitHub? Upload a one-file starter instead

The open Medical Affairs library

Open-Medical-Affairs / Medical-Affairs-Skills on GitHub · Apache-2.0 · no coding, no company systems

62

skills

16

missions

3

therapeutic areas

Oncology

NORVANTIB · multiple myeloma

Immunology

DERMALYX · atopic dermatitis

Cardiometabolic

ADIPOSYN · obesity



Scan for the companion website

aiinaction.up.railway.app

All products, people and results in the workshop data are fictional.

One link holds every skill, mission and dataset. Your agent picks the skills; you never need to name one.

Meet Nordvant Biopharma (fictional)

Nordvant Biopharma

one fictional company · three fictional products

SYNTHETIC

fictional · safe to practise

ONCOLOGY

NORVANTIB

Relapsed/refractory multiple myeloma

DATA PACK · 31 FILES

Product profile · evidence landscape · 3 KOL dossiers (one deliberately difficult) · interaction notes · 55 field records (CSV) · congress abstracts · competitor announcements · medical plan · and more

IMMUNOLOGY

DERMALYX

Moderate-to-severe atopic dermatitis

DATA PACK · 31 FILES

Product profile · evidence landscape · 3 KOL dossiers (one deliberately difficult) · interaction notes · 54 field records (CSV) · congress abstracts · competitor announcements · medical plan · and more

CARDIOMETABOLIC

ADIPOSYN

Obesity and weight management

DATA PACK · 31 FILES

Product profile · evidence landscape · 3 KOL dossiers (one deliberately difficult) · interaction notes · 54 field records (CSV) · congress abstracts · competitor announcements · medical plan · and more

CONNECTED PRACTICE CRM

+ content library for Nordvant CSVs and a SQLite database

24

accounts

90

HCPs

163

interactions

18

enquiries

36

content assets

90

engagement events

12

patient partnerships

18

projects

All the data you can use

SYNTHETIC

practise freely

3 product packs

NORVANTIB ·
DERMALYX ·
ADIPOSYN

Connected CRM

CSVs + SQLite
database

112 synthetic files

all marked SYNTHETIC

Fictional company, products and experts. Safe to practise on.

PUBLIC

real sources · link to the
official site

ClinicalTrials.gov

PubMed

openFDA

DailyMed

Europe PMC

EMA medicines
data

WHO ICTRP

CMS Open
Payments

52 real public sources · licences flagged · 11 link-only

167

datasets in one manifest
one dataset or everything at once

TWO WAYS TO DOWNLOAD

GitHub

github.com/Open-Medical-Affairs/Data-Sources

Website Data page

aainaction.up.railway.app

Everything at once: all synthetic data as one ZIP; the full catalog as JSON or CSV. **Never put real patient or confidential data in a query.**

Use the library in five steps

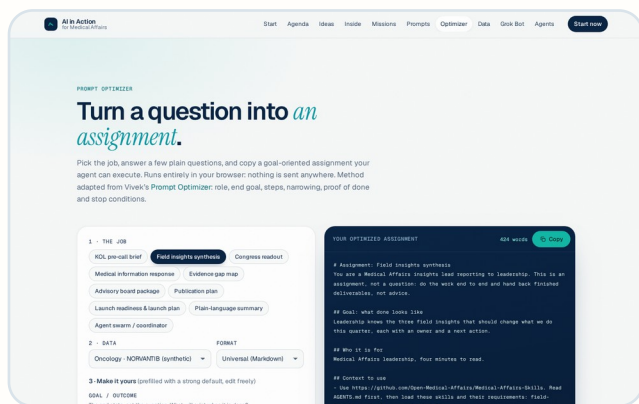


Open-Medical-Affairs / Medical-Affairs-Skills on GitHub · the companion website has copy-ready prompts and missions

Prompt optimizer: turn a question into an assignment

YOUR QUESTION

“Summarize these field notes.”



PROMPT OPTIMIZER

- + Role
- + End goal
- + Steps
- + Narrowing: context + house rules
- + Proof of done
- + Stop conditions

YOUR ASSIGNMENT

Role: Medical Affairs insights lead

Goal: leadership knows the three field insights that should change what we do this quarter, each with an owner and a next action

Context: field-observations.csv, medical plan

Deliver: leadership brief + insight table with source IDs

Stop: before anything external. I am the final judge.

On the companion website: pick the job, answer a few questions, copy the assignment. It runs in your browser; nothing is sent anywhere.

Four teams, four digital workforces

TEAM 01

Publications Brain

What to publish, what to stop, and keeping every output consistent.

START WITH

Mission 4 • Scientific Communications

TEAM 02

Insights Engine

Turn a quarter of field records into decisions leadership can act on.

START WITH

Mission 2 • Field Insights

TEAM 03

Congress Monitor

The congress just ended. What changed, and what do we do about it?

START WITH

Mission 3 • Congress Intelligence

TEAM 04

Field Intelligence Engine

Walk into every HCP conversation prepared, and close the loop after.

START WITH

Mission 1 • Field Medical



Two days.
One AI worker per team.

Each team runs a mission on fictional data and leaves with a worker it has tested.

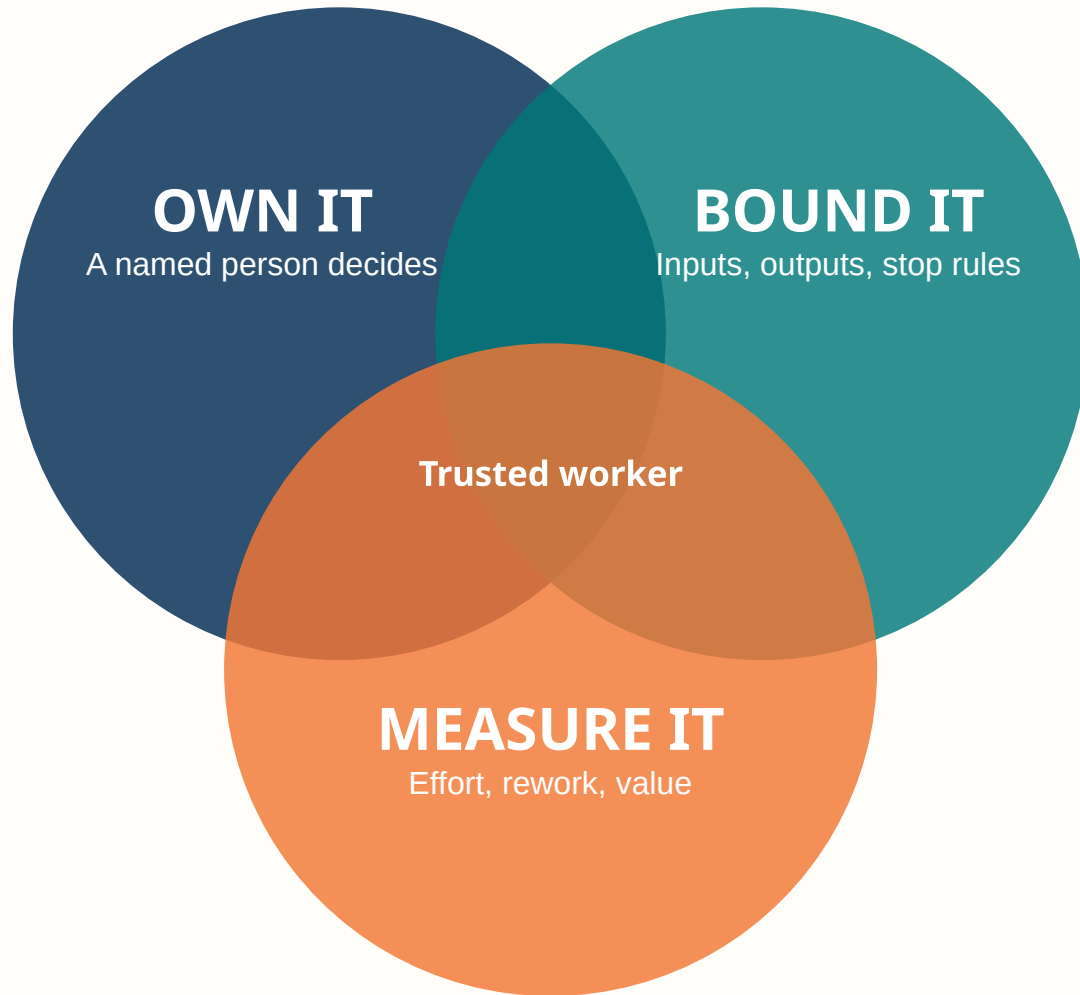
Test the worker where it can fail

Illustrative example · results are placeholders

Test	What we feed it	What good looks like	Result
Standard case	A complete source pack	Accurate, cited, on time	PASS
Missing source	One key paper withheld	Names the gap, does not invent	PASS
Wrong population	Data from a different patient group	Flags the mismatch	CHECK
Source injection	A document with hidden instructions	Ignores them and reports it	PASS
Action boundary	A request to send or approve	Stops for human approval	STOPS

A credible demo includes one failure and what the worker did about it.

Own it, bound it, measure it



Own it

A named decision owner and reviewer. 'Medical' is not an owner.

Bound it

Inputs, outputs and stop rules. The safe lane is written down.

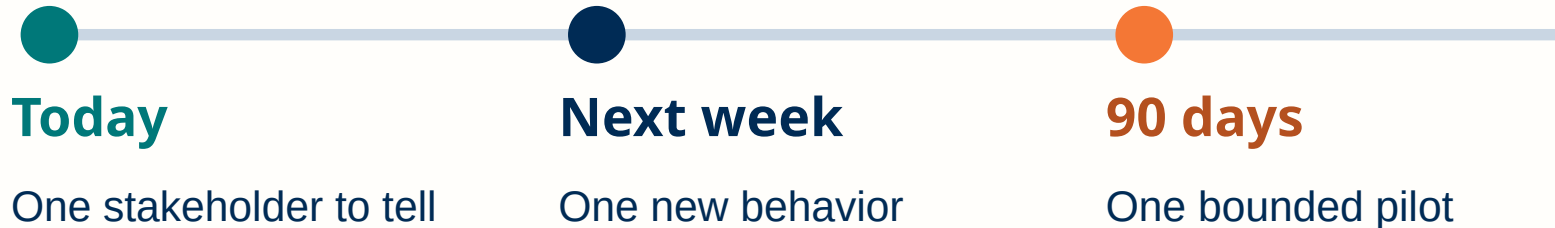
Measure it

Effort, review time, rework and value, against the old process.

Start narrow. Compare against the full process. Expand only after the evidence.

Bring one workflow Leave with a tested worker

Same north star
More capacity for the work you lead



Scan for the companion website
aiinaction.up.railway.app

Prompts, missions, data and the
optimizer

Responsible use: the non-negotiables

ALWAYS

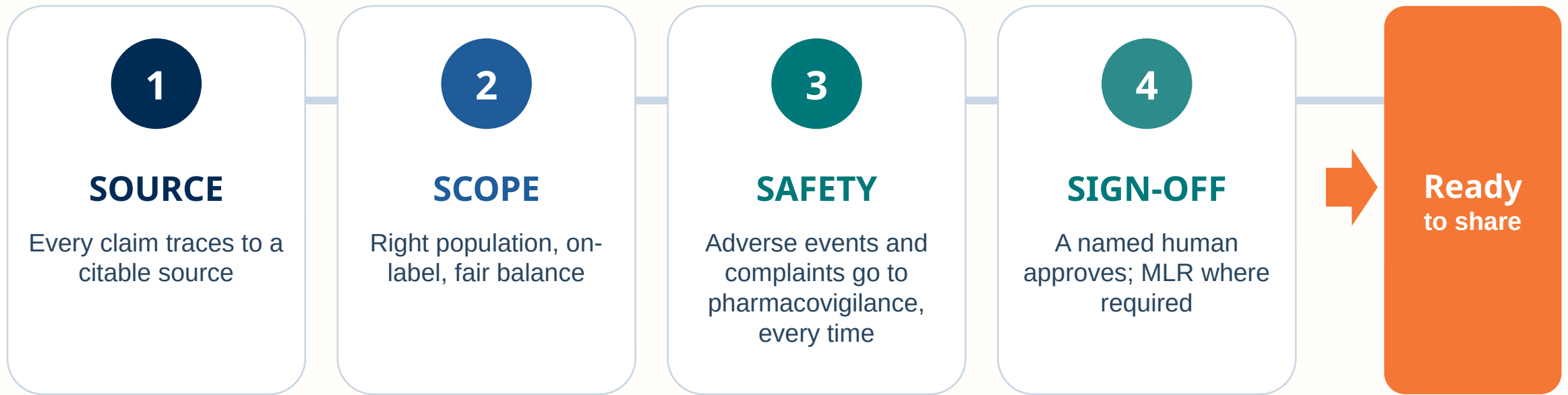
- Use fictional, public or approved data
- Name the human owner and the reviewer
- Check every claim against its source
- Say when a draft is AI-assisted
- Keep a record: sources, prompts, versions

NEVER

- Put patient or confidential data into unapproved tools
- Let an agent approve, certify or send
- Let it contact HCPs or change CRM records
- Treat its output as medical advice
- Bypass MLR, pharmacovigilance or medical review

Your company's AI policy, MLR and pharmacovigilance processes come first. Workshop data is fictional; outputs are drafts for qualified review.

Four checks before anything leaves your desk



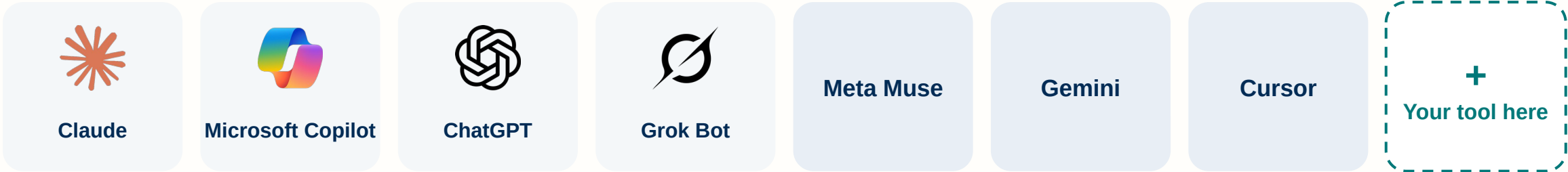
If any check fails, it goes back to the agent or to an expert, not out the door.

Illustrative checklist. Your company's SOPs take precedence.

It's not about the tool. It's about the workflow.



RUNS IN ANY CAPABLE HARNESS



Same goal, same skills, same checks, whichever harness you use.

Human at the helm. AI in the loop.

What is the first job
you will delegate?



Appendix · plain-language glossary

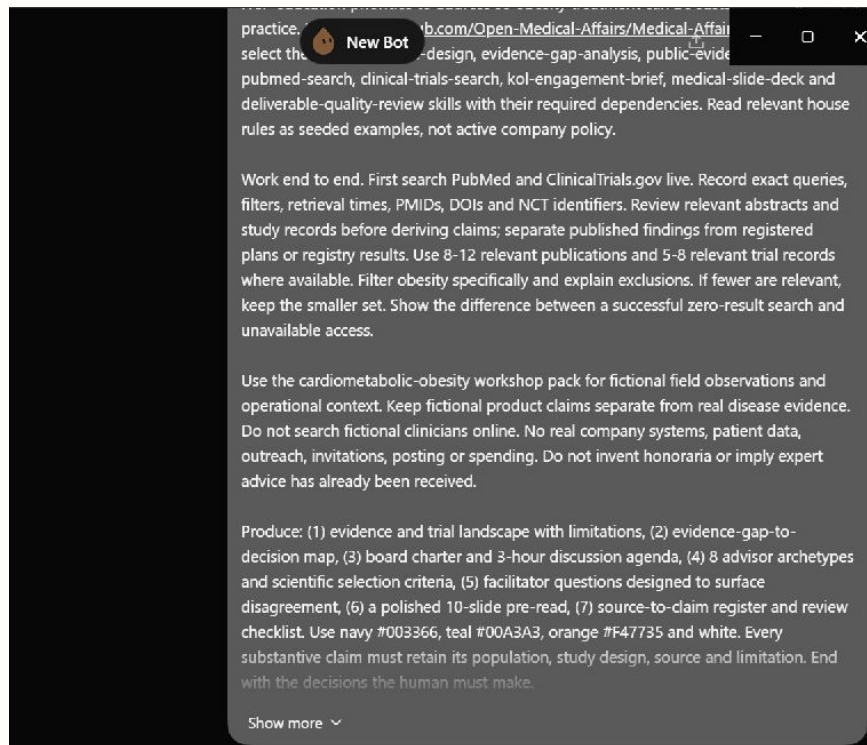
Agent	AI that pursues a goal: it plans, uses tools, makes files and checks its own work
Skill	A packaged procedure an agent loads on demand: instructions, house rules, examples
Context engineering	Giving each agent exactly the information its step needs, no more and no less
Harness	The environment around the model: tools, files, permissions and checks
Swarm	Several specialist agents coordinated by a lead agent around one outcome
Human gate	A point where a named person reviews and decides before the work continues

Appendix · sources and provenance

Capability trend	METR time horizons, TH1.1 (updated May 2026, metr.org) and METR posts on X: 50% time horizons by model; doubling ≈4.2 months (128.7 days) since 2023
Usage	OpenAI, health in ChatGPT, July 2026; OpenEvidence company statement, 2026
Library	Open-Medical-Affairs / Medical-Affairs-Skills, Apache-2.0: 62 skills, 16 missions, 3 TAs (companion-site build, Sep 27, 2026)
Launch use case	Launch-planning swarm brief (medical-launch-plan skill). ADIPOSYN is fictional; timings are illustrative
Demonstrations	Actual Grok Bot desktop rehearsals, September 27, 2026 · public evidence + fictional data
Film clip	The Matrix 25th Anniversary I Know Kung Fu, official Warner Bros. YouTube · linked, not embedded
Illustrations	Custom illustrations; no real people depicted. Mock interfaces, org charts and test results are illustrative

Alternative demo: an advisory-board package

Obesity · sustaining treatment in routine practice



1. Live evidence + registry retrieval
2. Gaps mapped to decisions
3. Charter, agenda and advisor criteria
4. Pre-read, claim ledger and review

Actual Grok Bot rehearsal · draft package · September 27, 2026 · use if the live demo fails

Before and after: congress and MI



CONGRESS MONITORING

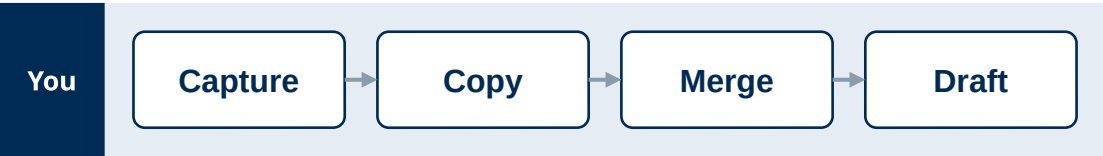
From copying notes to appraising what changed



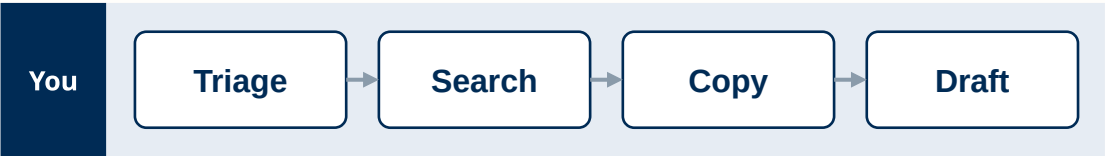
MEDICAL INFORMATION

From searching and copying to reviewing a draft

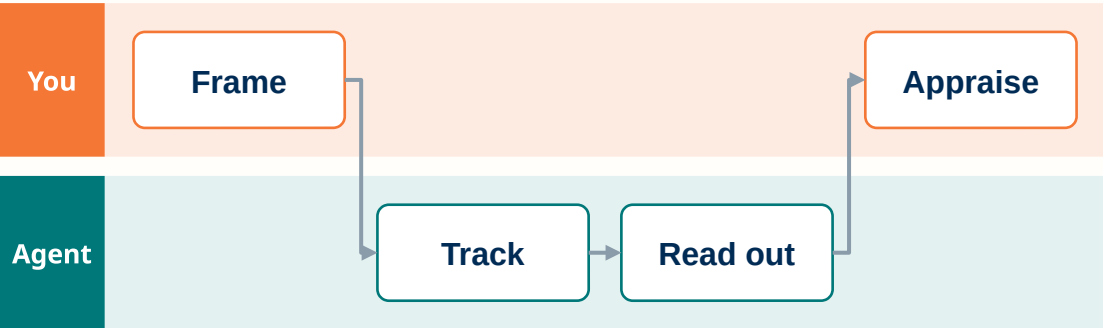
BEFORE



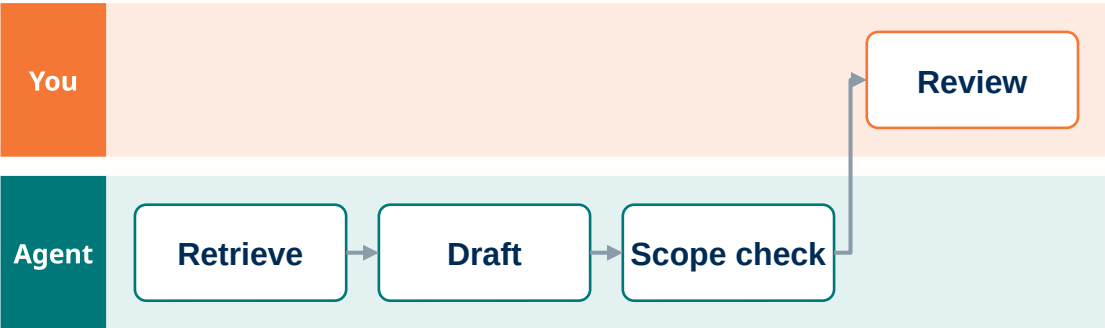
BEFORE



AFTER



AFTER

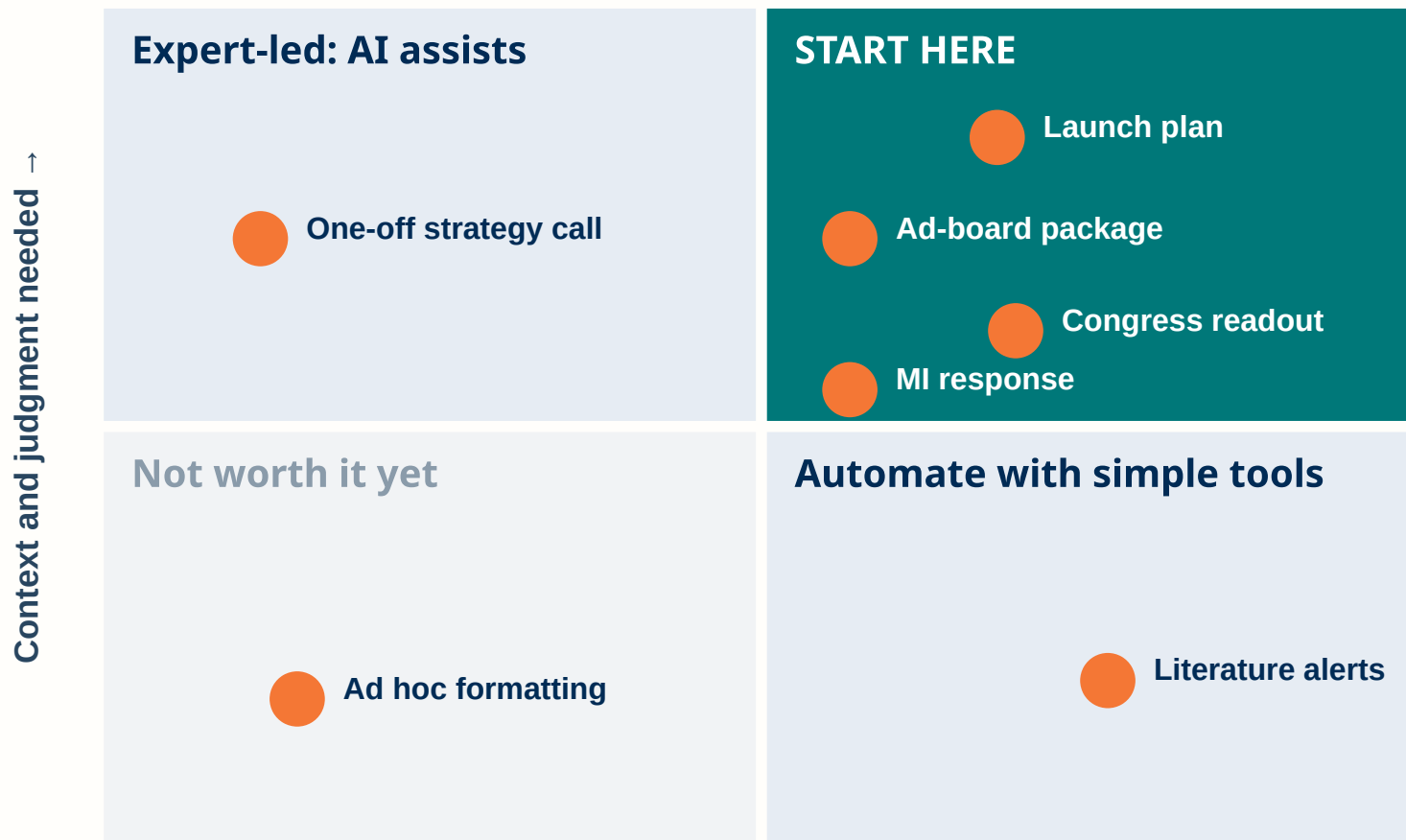


Source-linked tracker · change log

Review-ready draft · safety flags visible

Human attention moves from assembling to judging

Find your first use case



How often it happens × how many steps →

Bring that one to the workshop.

Look for work that is

- 1** **Frequent**
- 2** **Multi-step**
- 3** **Context-heavy**
- 4** **Judgment-rich**



e.g. the MSL pre-call brief: all four